

JS Object & Array Destructuring - Practice Questions

1. Copy an Array Using Spread Operator

Question:

Write a function `copyArray` that creates a copy of an array using the spread operator.

Input:

```
const numbers = [1, 2, 3, 4, 5];
```

Expected Output:

```
[1, 2, 3, 4, 5]
```

2. Merge Two Arrays Using Spread Operator

Question:

Write a function that merges two arrays using the spread operator.

Input:

```
const arr1 = [1, 2, 3];  
const arr2 = [4, 5, 6];
```

Expected Output:

```
[1, 2, 3, 4, 5, 6]
```

3. Copy an Object Using Spread Operator

Question:

Write a function that creates a copy of an object using the spread operator.

Input:

```
const person = { name: "Alice", age: 25 };
```

Expected Output:

```
{ name: "Alice", age: 25 }
```

4. Merge Two Objects Using Spread Operator

Question:

Write a function that merges two objects using the spread operator.

Input:

```
const obj1 = { a: 1, b: 2 };  
const obj2 = { c: 3, d: 4 };
```

Expected Output:

```
{ a: 1, b: 2, c: 3, d: 4 }
```

5. Use Rest Operator to Collect Arguments

Question:

Write a function that takes multiple numbers and returns their sum using the rest operator.

Input:

```
console.log(sum(2, 3, 5));
```

Expected Output:

```
10
```

6. Extract First Two Elements Using Array Destructuring

Question:

Write a function that extracts the first two elements of an array using destructuring.

Input:

```
const arr = [10, 20, 30, 40];
```

Expected Output:

```
[10, 20]
```

7. Swap Two Variables Using Array Destructuring

Question:

Write a function that swaps two variables using destructuring.

Input:

```
let a = 5, b = 10;
```

Expected Output:

```
[10, 5]
```

8. Extract Specific Properties Using Object Destructuring

Question:

Write a function that extracts `name` and `age` from an object using destructuring.

Input:

```
const user = { name: "Bob", age: 30, city: "NYC" };
```

Expected Output:

```
{ name: "Bob", age: 30 }
```

9. Default Values in Object Destructuring

Question:

Write a function that provides a default value if a property is missing.

Input:

```
const user = { name: "Charlie" };
```

Expected Output:

```
{ name: "Charlie", age: 25 } // Default age is 25
```

10. Rest Operator in Object Destructuring

Question:

Write a function that extracts `name` and stores the rest of the properties using the rest operator.

Input:

```
const user = { name: "David", age: 40, city: "LA", job: "Engineer" };
```

Expected Output:

```
{ name: "David", others: { age: 40, city: "LA", job: "Engineer" } }
```

JS Object & Array Destructuring - Assignment Questions

11. Clone an Object Without Modifying the Original

Question:

Write a function that clones an object and updates only one property.

Input:

```
const person = { name: "Eve", age: 35 };
```

Expected Output:

```
{ name: "Eve", age: 40 }
```

12. Convert Function Parameters into an Object Using Rest Operator

Question:

Write a function that accepts multiple parameters and stores them as an object.

Input:

```
console.log(storeUserData("John", 28, "Developer"));
```

Expected Output:

```
{ name: "John", age: 28, job: "Developer" }
```

13. Remove Duplicates from an Array Using Spread and Set

Question:

Write a function that removes duplicates from an array.

Input:

```
const arr = [1, 2, 2, 3, 4, 4, 5];
```

Expected Output

```
[1, 2, 3, 4, 5]
```

14. Combine Multiple Arrays Using Spread

Question:

Write a function that merges three arrays into one.

Input:

```
const a1 = [1, 2], a2 = [3, 4], a3 = [5, 6];
```

Expected Output:

```
[1, 2, 3, 4, 5, 6]
```

15. Dynamic Key Assignment Using Spread

Question:

Write a function that dynamically assigns a key to an object.

Input:

```
console.log(createObject("price", 500));
```

Expected Output:

```
{ price: 500 }
```